



RAMCO INSTITUTE OF TECHNOLOGY

(An Autonomous Institution)

Approved by AICTE, New Delhi & Affiliated to Anna University
NAAC Accredited with 'A+' Grade & An ISO 9001: 2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE, MECH and CIVIL

Department of Information Technology Academic Year 2025 - 2026 (Even Semester)

Semester, Degree & Branch: II Semester, B.Tech & IT

Course Code & Title: IT25201& Foundations of Data Science using Python

Name of the Faculty member (s): Mrs. P.Ramya, AP/IT

Innovative Practice Description

Unit / Topic: Module V / Data Aggregation and Group Operations: Group By Mechanics

Course Outcome: CO5

Topic Learning Outcome: -

Activity Chosen: Flipped Classroom

Justification:

The Flipped Classroom model is ideal for GroupBy Mechanics because it shifts the focus from memorizing code syntax to mastering the logical Split-Apply-Combine framework through active practice. Since grouping operations often involve complex multi-step reasoning and frequent debugging of indexing errors, students benefit most from tackling these hurdles in an interactive environment where they can receive immediate instructor feedback. By shifting basic lectures to pre-class videos, the actual session transforms into a laboratory for data intuition, allowing students to debate aggregation strategies and solve real-world data problems collaboratively rather than struggling with technical roadblocks alone at home.

Time Allotted for the Activity: 15 Minutes

Details of the Implementation:

- Following the flipped classroom model, V.P.Vaishnavi was randomly selected to lead the presentation based on the preparatory materials previously uploaded to the LMS.
- The presenter provided a detailed breakdown of the Split-Apply-Combine framework, illustrating how raw data is partitioned and transformed into meaningful summaries.
- After the explanation, V.P.Vaishnavi initiated a knowledge check by questioning the class. This sparked a dynamic exchange where some students provided answers while others raised technical doubts regarding specific grouping functions.
- The entire group participated in a focused discussion to resolve these queries, allowing for a deeper collective understanding of the mechanics.
- The activity concluded with the summarizing the key logic of the topic and finalizing the assigned task, with the resulting outputs documented in Figures 1.

CO – PO / PSO mapping:

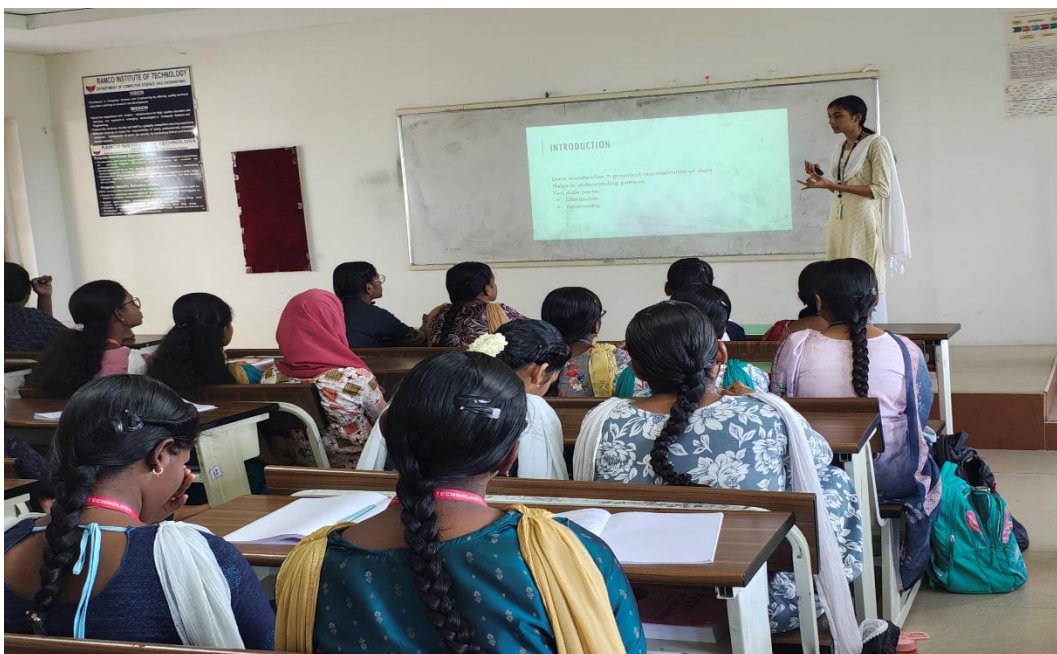
CO	PO1	PO2	PO5	PO10	PSO1	PSO2
CO5	3	3	2	1	3	2

(1 – Low 2 – Moderate 3 – High)

PO/PSO Mapped:

Innovative Practice	PO1	PO2	PO5	PO10	PSO1	PSO2
	3	3	2	1	3	2
Justification for Correlation	Applying mathematical and algorithmic principles to transform raw data into aggregated summaries.	Identifying how to split complex datasets into logical groups to reach substantiated conclusions.	Selecting and applying appropriate tools like Pandas, SQL, or R to execute grouping operations.	Clearly explaining complex data mechanics and responding to peer doubts during the 15-minute presentation.	The ability to analyze large datasets and extract meaningful insights through aggregation techniques to support business intelligence.	Demonstrating expertise in using libraries and frameworks to implement Split-Apply-Combine patterns for real-world software applications.

- Images / Screenshot of the practice:**

**Figure 1. V.P.Vaishnavi Presenting the topic****Reflective Critique:**❖ **Feedback of practice from students and other stakeholders:**

- Students found that the flipped classroom approach made the logical "Split-Apply-Combine"

framework much easier to visualize and understand.

- Learners reported that using real-world datasets (like global sales or weather patterns) improved their grasp of how aggregation is used in professional data roles.
- Students appreciated the peer discussions, noting that debugging complex grouping errors together increased their technical confidence and teamwork skills.

❖ ***Benefit of the practice:***

- Students improved their data manipulation accuracy and coding speed through guided, hands-on practice with live datasets.
- The activity simulates industry expectations by challenging students to handle messy, real-world data with multiple grouping keys.
- By resolving technical "KeyErrors" in class, students developed greater independence in applying advanced Pandas or SQL grouping concepts to practical problems.

❖ ***Challenges faced in implementation:***

- Students with weaker Python or SQL fundamentals required extra individualized support to grasp the logic of multi-index grouping.
- Uneven participation occurred when some students skipped the pre-class LMS videos, causing a knowledge gap during the collaborative group tasks.
- Time management proved difficult as the 15-minute window was often stretched by deep-dive discussions into specific data edge cases.

References:

https://www.ritrjpm.ac.in/images/B.Tech-IT/2025-2026/GSS_CCS370_Flipped_Classroom_U3.pdf

Signature of Faculty Member

HOD